

Unit 5, Lesson 1: Solving Multi-Step Equations



Benchmark

MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.



Learning Target

☐ I can solve multi-step linear equations.



5.1.1 Warm-Up: Aviation Careers

Aviation careers include pilots, air-traffic controllers, airport operators, airfield maintenance, aircraft rescue and firefighting, law enforcement and security officers, and project management.



1. Which of these careers in aviation had you not heard of before?
2. Have you ever thought about a career in aviation?



5.1.2 Refresher: Solving Two-Step Equations

1. Work with your partner to complete the statements using the words from the bank. Words may be used more than once.

coefficient

constant

variable

- a. In the expression $7x - 2$, x is the _____,
7 is a _____, and -2 is a
_____.
 - b. When solving an equation, the goal is to isolate the
_____.
2. Solve each equation and verify your solution using substitution.
 - a. $8.25 - 1.5y = 1.8$
 - b. $21 = 36\left(\frac{2}{3}a + \frac{5}{4}\right)$

3. Compare your work with your partner's. Summarize any similarities and/or differences in your solution methods.



5.1.3 Collaboration: Solving Multi-Step Equations

Javier and Arnav solved the same two equations, but they used different first steps.

1. In the table, **each student's first step** for solving each equation is started **in blue**. Work with your partner to finish solving the equations, continuing from each student's first step.

Javier	Arnav
$47 - 5.7m = 2.3m + 52$ $+ 5.7m \quad + 5.7m$	$47 - 5.7m = 2.3m + 52$ $- 47 \qquad \qquad - 47$
$-\frac{1}{4}(28p + 60) = -6p + 1$ $\left(-\frac{1}{4}\right)(28p) + \left(-\frac{1}{4}\right)(60) = -6p + 1$	$\frac{-\frac{1}{4}(28p+60)}{-\frac{1}{4}} = \frac{-6p+1}{-\frac{1}{4}}$

2. Discuss with your partner why both students had the same solutions for each equation, even though they used different strategies.
3. Discuss which strategies you preferred to start with when solving each of these equations. Summarize your discussion.



5.1.4 Guided Practice: Solving Multi-Step Equations

1. Solve each equation and verify the solution using substitution.

a. $8 + 5(16 + 9g) = 35g$

b. $49 - 6d = \frac{3}{2}(28d - 10)$

c. $\frac{62k-27}{11} = 8k + 7$



5.1.5 Your Turn!

1. Solve each equation.

a. $4.7y + 1.6 - 0.2y = 6.6$

b. $\frac{8}{3} + 29z + 1 = 7z - \frac{1}{3}$



5.1.6 Wrap-Up: Cool Down

1. Solve each equation and check the solution using substitution.

a. $23.5 - 32p = 16.5p + 4.1$

b. $\frac{3}{4}(12n - 10) = 19n - 3.5$



Unit 5, Lesson 2: Solving Equations and Rearranging Formulas



Benchmarks

MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.



Learning Targets

- ☐ I can solve multi-step linear equations.
- ☐ I can isolate a quantity of interest in a formula by rearranging a formula.